MIT | Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	COURSE SYLLABUS	
Department of Computer Science and Engineering(AIML)	**W.E.F**	**AY:** 2026 - 2027 (Rev. 2023 (NEP))
THIRD YEAR BACHELOR OF TECHNOLOGY CSE(AIML)	**COURSE NAME**	GENERATIVE AI
	COURSE CODE	2311312T
	COURSE CREDITS	2
RELEASE DATE : 01/07/2026	**REVISION NO.**	0.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
2	NIL	15	20	40	NIL	NIL	75

PREREQUISITE KNOWLEDGE: Linear Algebra, Probability and Statistics, Machine Learning, Artificial Intelligence, Python programming

COURSE OBJECTIVES :

2311312T.CEO.1: To introduce the principles of deep neural networks and convolutional architectures for visual perception tasks.

2311312T.CEO.2: To familiarize learners with sequence modeling techniques and core natural language processing concepts.

2311312T.CEO.3: To develop an understanding of unsupervised representation learning and generative latent-variable models.

2311312T.CEO.4: To explain the foundations and training dynamics of modern generative models, including adversarial and diffusion-based approaches.

2311312T.CEO.5: To impart conceptual knowledge of large language models, retrieval-augmented systems, and multimodal Generative AI applications.

COURSE OUTCOMES :

After successful completion of the course, students will be able to,

2311312T.CO.1: Apply deep learning and computer vision techniques for image analysis and object detection. [L3]

2311312T.CO.2: Apply sequence models and NLP techniques for language processing applications. [L3]

2311312T.CO.3: Compare representation learning and generative models for feature learning and data generation. [L4]

2311312T.CO.4: Analyze diffusion models and Large Language Models for Generative AI applications. [L4]

2311312T.CO.5: Evaluate Generative AI solutions using prompt engineering, RAG, and multimodal AI techniques.[L5]

COURSE ABSTRACT:

This course introduces the fundamentals of Generative AI, covering deep learning, computer vision, natural language processing, representation learning, and modern generative models. Students will explore convolutional neural networks (CNNs), sequence models, autoencoders, Generative Adversarial Networks (GANs), Diffusion Models, and transformer-based Large Language Models (LLMs). The course also covers prompt engineering, Retrieval-Augmented Generation (RAG), multimodal AI, and responsible AI practices through real-world applications and case studies, preparing students for emerging AI technologies and applications.

THEORY COURSE CONTENTS

UNIT 1	**Deep Learning Foundations and Computer Vision**	**06 HOURS**

Applications/Case Study: Handwritten Digit Recognition using MLP (MNIST dataset)
Contents: Introduction to Deep Learning, Artificial Neural Networks and Perceptron, Forward and Backpropagation, Loss Functions (MSE, Cross Entropy), Activation Functions (Sigmoid, Tanh, ReLU, Leaky ReLU), Optimizers (SGD, Adam) Regularization (Dropout, Batch Normalization),CNN Architecture, Transfer Learning, Object Detection Fundamentals, YOLOv8, Performance Metrics (IoU, Precision, Recall, mAP)
Self Study: Perform object detection using YOLOv8 on sample images.
Further Readings: Applications of computer vision in healthcare and smart cities

UNIT 2	**Natural Language Processing**	**05 HOURS**

Applications/Case Study: Sentiment analysis of product reviews
Contents: Text Preprocessing, Bag of Words, TF-IDF, Word Embeddings (Word2Vec, FastText), Recurrent Neural Networks (RNN), Vanishing Gradient Problem, Long Short-Term Memory (LSTM), Sequence-to-Sequence Learning, Introduction to Attention Mechanism
Self Study: Perform text classification using pre-trained word embeddings.
Further Readings: Attention mechanism in NLP and Applications of NLP in conversational AI

UNIT 3	**Representation Learning and Autoencoders**	**04 HOURS**

Applications/Case Study: Image denoising using Autoencoders
Contents: Representation Learning, Dimensionality Reduction using PCA, Autoencoders, Denoising Autoencoders, Variational Autoencoders, Latent Space Representation, bridging representations into generative modeling
Self Study: Visualize latent representations of handwritten digits.
Further Readings: Variational Autoencoders for image generation

UNIT 4	**Generative Adversarial Networks**	**05 HOURS**

Applications/Case Study: Medical image augmentation
Contents: Introduction to Generative AI, GAN Architecture, Generator and Discriminator, Deep Convolutional GAN (DCGAN), Conditional GAN (cGAN), StyleGAN Overview, Applications of GANs
Self Study: Generate handwritten digits using GANs.
Further Readings: :Ethical concerns in synthetic media generation

UNIT 5	**Diffusion Models**	**05 HOURS**

Applications/Case Study: AI-generated digital artwork

Contents: Introduction to Diffusion Models, Forward and Reverse Diffusion Process, Denoising Diffusion, Probabilistic Models (DDPM) – Concept, Stable Diffusion, Prompt-based Image Generation, Image Editing using Diffusion Models, Applications of Diffusion Models

Self Study: Generate images using Stable Diffusion with different prompts.

Further Readings: :Advances in diffusion-based generative models, Text-to-image generation techniques

UNIT 6	Large Language Models and Modern Generative AI	05 HOURS

Applications/Case Study: Agricultural advisory chatbot using RAG

Contents: Transformer Architecture, Self-Attention Mechanism, Encoder-only, Decoder-only and Encoder–Decoder Models, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases, Multimodal Generative AI, AI Agents (Introduction), Responsible AI, Hallucination, Bias and Safety

Self Study: Build a simple RAG-based question-answering system.

Further Readings: :Responsible AI, AI ethics, and governance and Emerging trends in Large Language Models

TEXT BOOKS

1. Goodfellow, Bengio, and Courville(2024). *Deep Learning.* 1st Edition, The MIT Press. ISBN 978-0262337373.
2. David Foster(2023). *Generative Deep Learning.* 2nd Edition, Shroff/O'Reilly. ISBN 978-9355429988.
3. Daniel Jurafsky and James H. Martin (2020). *Speech and Language Processing.* 2nd Edition, Pearson. ISBN 978-9332518414.

REFERENCE BOOKS

1. Seth Weidman (2019). *Deep Learning from Scratch: Building with Python from First Principles.* 1st Edition, O'Reilly Media.ISBN 978-9352139026.
2. Aurélien Géron (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.* 3rd Edition, O'Reilly Media. ISBN 978-9355421982.
3. Sebastian Raschka (2024). *Build a Large Language Model from Scratch.* 1st Edition, Manning Pubns Co.ISBN 978-1633437166.
4. Lewis Tunstall, Leondro von Werra, Thomas Wolf (2022). *Natural Language Processing with Transformers.* 1st Edition, Shroff/O'Reilly, ISBN 978-9355420329.

E-RESOURCES

1. https://www.deeplearningbook.org/
2. https://builtin.com/machine-learning/deep-learning
3. Ultralytics YOLO Documentation – https://docs.ultralytics.com
4. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26/cs181
5. Hugging Face Documentation – https://huggingface.co/docs
6. Stanford CS224N – https://web.stanford.edu/class/cs224n/